

INTEGRATION TEST REPORT

TC	Test condition	Expected Outcome	Observed Outcome	Status (pass or fail)
T1. Normal Start	Mode: ACC -> IGNITION_ON -> IGNITION_ON [S:40 T:80 G:1]	No faults, NORMAL state	Mode=IGNITION_ON, State=NORMAL, Faults=NONE	PASS
T2. Overspeed Detection	Mode: IGNITION_ON [S:130 T:85 G:4]	Overspeed fault set, fault counter incremented, DEGRADED state	Mode=IGNITION_ON, State=DEGRADED, Faults=OVERSPEED	PASS
T3. High Temperature Warning	Mode: IGNITION_ON [S:60 T:100 G:3]	High temperature warning raised, no crash, correct priority handling	Mode=IGNITION_ON, State=DEGRADED, Faults=HIGH_TEMP	PASS
T4. Critical Overheat	Mode: IGNITION_ON [S:60 T:115 G:3]	Critical fault raised, state escalated, fault reason logged	Mode=IGNITION_ON, State=DEGRADED, Faults=CRIT_OVERHEAT	PASS
T5. Invalid Gear	Mode: IGNITION_ON [S:60 T:80 G:0]	Invalid gear fault raised, safe handling, no crash	Mode=IGNITION_ON, State=DEGRADED, Faults=INV_GEAR	PASS
T6. Illegal Mode Transition	Mode: IGNITION_ON [S:0 T:25 G:0]	Illegal transition detected, FAULT mode triggered	Mode=FAULT, State=DEGRADED, Faults=ILL_MODE	PASS
T7. Multiple Faults in One Cycle	Mode: IGNITION_ON [S:140 T:120 G:0]	All three faults set, highest priority respected, state escalated to SAFE	Mode=FAULT, State=SAFE, Faults=OVERSPEED, CRIT_OVERHEAT, INV_GEAR	PASS
T8. Persistent Fault Across Cycles	Mode: IGNITION_ON -> IGNITION_ON -> IGNITION_ON [S:60 T:115 G:3]	Fault counter increments each cycle, SAFE state reached	Mode=FAULT, State=SAFE, Faults=CRIT_OVERHEAT, ILL_MODE	PASS
T9. Recovery and Reset Logic	Mode: IGNITION_ON -> IGNITION_ON [S:40 T:80 G:2]	Without OFF reset, SAFE latch remains and state does not return to NORMAL	Mode=FAULT, State=SAFE, Faults=ILL_MODE	PASS